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Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics Engineering

Lesson title:

Robotics and Automation module

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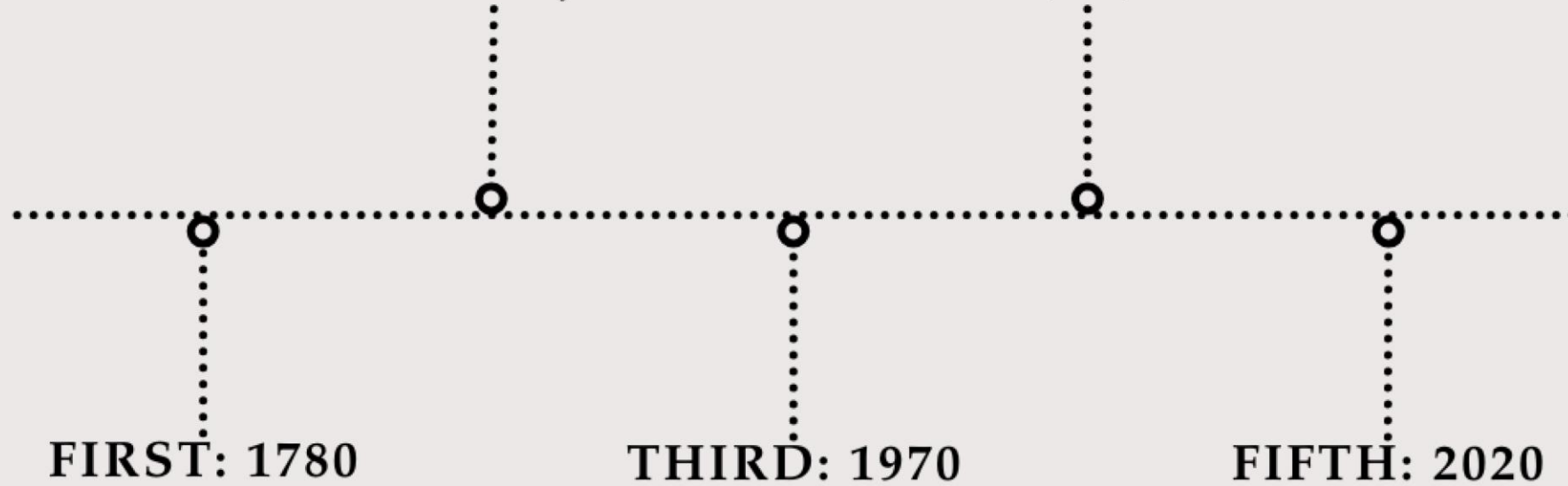
INDUSTRIAL REVOLUTIONS

SECOND: 1900

Key technologies:
mass production,
assembly lines,
electricity

FOURTH: 2010

Key technologies:
IoT, IIoT, Cyber-
Physical Systems
(CPS)



Key technologies:
machinery, water
and steam power

Key technologies:
computers,
automation,
electronics

Key technologies:
Human-Centric CPS
(HCPS), Cobots, AI
& Machine Learning

Industrial Automation

Industrial automation involves technologies that use **mechanical, electronic, and IT systems** for **control and production** in industries. It is also called **production process automation**.

Automation can be classified based on **production methodologies**:

Rigid Automation

Process Characteristics:

- Mass production of large batches with **constant product characteristics**
- **Fixed sequences** of operations executed repeatedly

Machine Characteristics:

- **Single-purpose, highly specialized mechanical design**, optimized for one type of work to maximize production efficiency
- **Dedicated control system**, implemented via **PLC or hardwired techniques**, with **limited or no reprogramming capability** for other tasks

Industrial Automation

Programmable Automation

Process Characteristics:

- Production of **small to medium batches** with **variable product characteristics**
- **Multiple sequences of diverse operations**

Machine Characteristics:

- **Versatile mechanical architecture**, designed to operate on **different items within the same technology group**
- **Reprogrammable and/or adaptive control system**

Industrial Automation

Flexible Automation

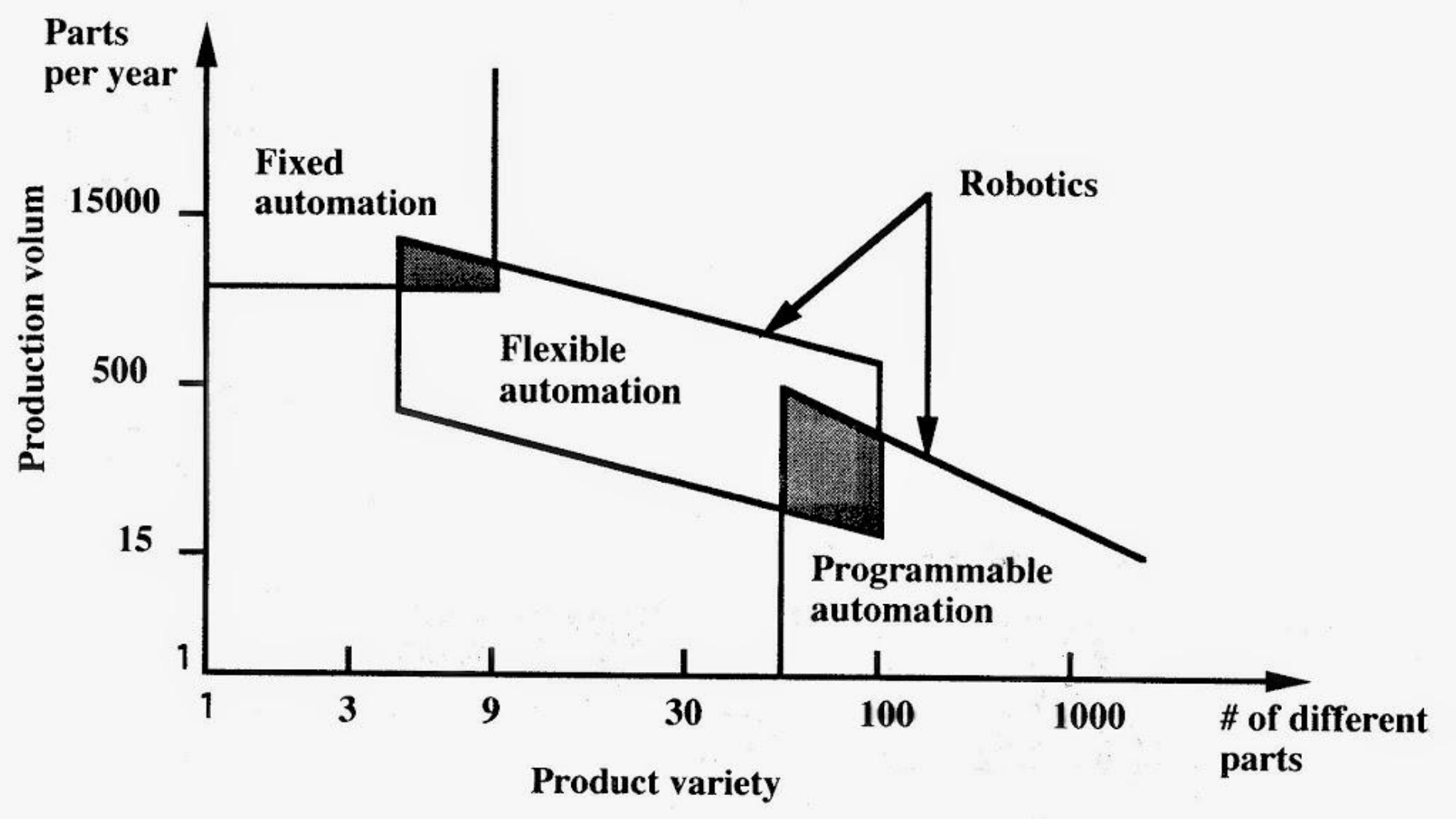
Process Characteristics:

- **On-demand production** of batches with **variable sizes** and **different products**
- Requirement for **high production rates**
- Operations requiring **sophisticated technologies**
- **Internal and warehouse logistics**

Machine Characteristics:

- **Highly versatile mechanical architecture**, designed to handle **different items with diverse technologies**
- **Reprogrammable control system**
- **High integration with the company's information system**

Different types of automation (indicative values)



What tasks should a robot perform?

Handling (or transportation):

This category includes all applications in which the robot does not directly perform an operation on the manufactured parts, but rather moves them between the machines operating within a processing cell.

- Palletizing: arranging objects in a predefined order on a pallet.
- Loading and unloading of warehouses.
- Loading and unloading of operating machines and machine tools.
- Etc.



What tasks should a robot perform?

Manipulation:

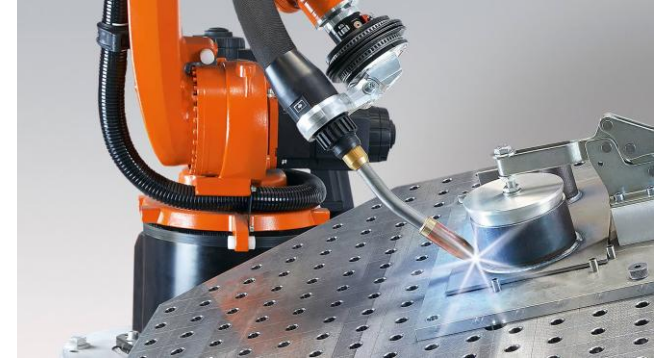
Divided into two subcategories:

- **Processing**

This category includes all operations the robot performs on products **using a tool**.

Examples of operations:

- Arc and spot **welding**.
- Cutting.
- Painting.
- Milling and drilling, etc.



- **Assembly**

Examples of operations include:

- Assembly of mechanical and electrical components.
- Mounting of electronic boards.
- Screwing/fastening.



What tasks should a robot perform?

Measurement:

Some typical measurement tasks include:

- Profile detection.
- Dimensional testing.
- Inspection for product quality assessment and defect detection.

To perform these tasks, the robot is usually assisted by dedicated machinery (automatic or semi-automatic), such as:

- CNC cutting machines
- Material handling systems
- Tool-changing equipment

The set of machines used in a process or group of subprocesses forms the so-called **work cell**.

Work cell

Efficient organization of the robotic cell is of **fundamental importance**.

The most common configurations include:

Central Robot Cell: The **robot is positioned at the center** and surrounded by processing machines, which it loads and unloads.

Production Line: One or more **robots are placed along a conveyor line**. The line transports the workpieces, and each robot performs a specific operation or assembly task.

Mobile Robot Cell: The robot arm is mounted on a **mobile platform** that moves to different positions within the cell to perform various operations. Ideal when a single robot must serve multiple machines.

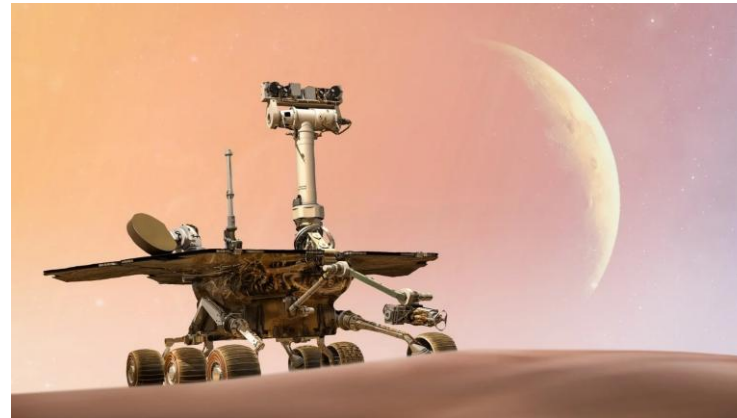
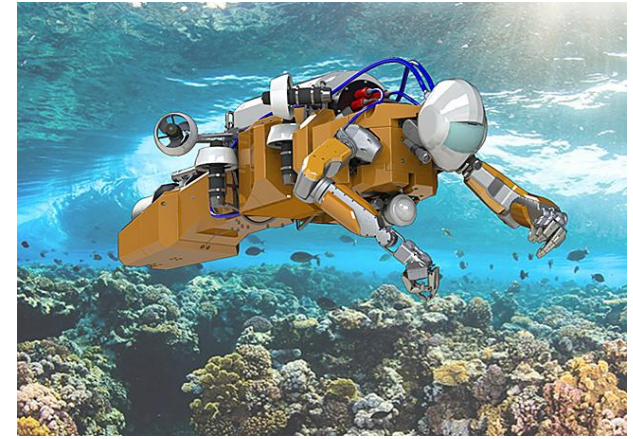
Autonomous Guided Vehicles (AGVs): **Mobile robots** designed to transport parts and/or tools within a large flexible manufacturing cell containing multiple CNC machines and storage systems. Typically, these robots handle transportation and precise positioning, not manipulation. Advanced implementations of this type are often referred to as “automated factories.”

Service Automation

Exploration Robotics

Some examples:

- **Underwater** (submarine exploration).
- **Space** (planetary or orbital missions).
- **Volcanic** (extreme-environment research).



Service Automation

Home Automation (Domotics)

Examples:

- Robotic **lawn mowers**.
- Robotic **vacuum cleaners**, etc.



Service Automation

Biomedical Applications

- Rehabilitation robotics.
- Surgical systems.
- Teleoperation and remote procedures.



Service Automation

Training, Entertainment & Gaming

- Haptic interfaces.
- Virtual reality.
- Telepresence systems.



Elements of a Robot

General Characteristics

A robot generally consists of:

- **Mechanical Structure** – the physical framework or body of the robot.
- **Actuators** – devices that produce motion (e.g., motors, hydraulic systems).
- **Sensors** – components that detect environmental or internal conditions.
- **Control System** – processes information and coordinates the robot's actions.

Elements of a Robot

Actuators

Actuators can be divided into **three main categories**:

- **Electrical Actuators**

(DC motors, servos, stepper systems, solenoids)

- **Hydraulic Actuators**

(use pressurized fluids for high force applications)

- **Pneumatic Actuators**

(use compressed air for fast, lightweight motion)

Elements of a Robot

Electric Actuators (~50% of cases)

Advantages:

- High speed and precision.
- Compatible with advanced control algorithms.
- Easy to find and low cost.
- Simple to use.
- Compact and lightweight.

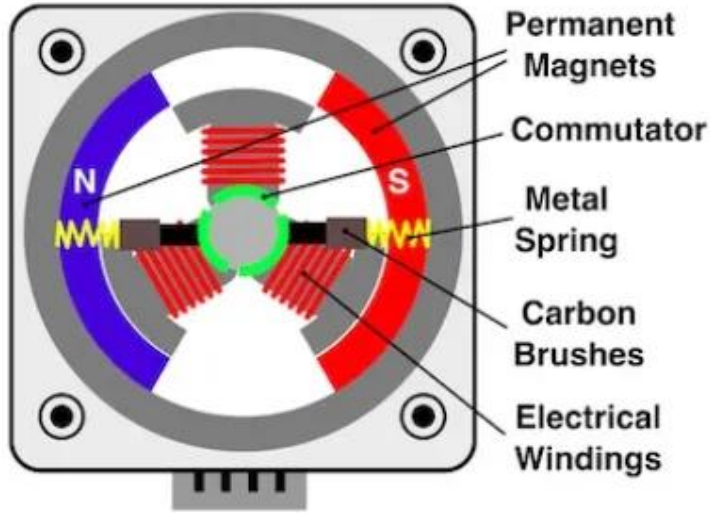
Disadvantages:

- Can require a gearbox (this produce reduced of precision, bulkier, higher cost).
- Limited available power.

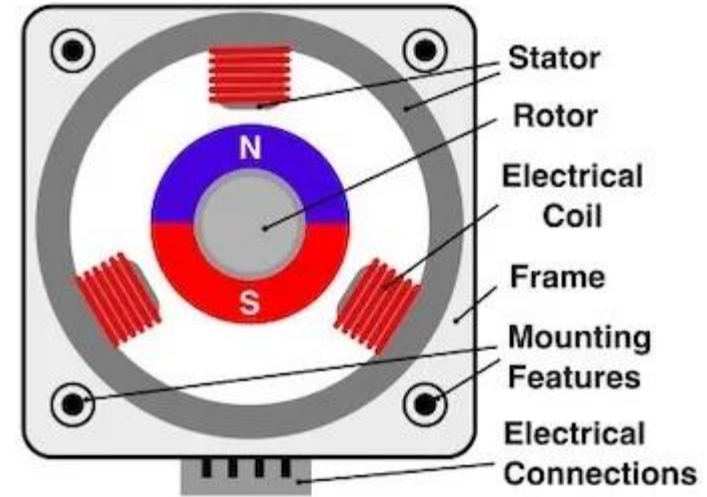


Electric motors (DC)

Main components



Brushed motor



Brushless motor

In brushless motors, the copper windings have been moved from the rotor to the stator, and the permanent magnet is now part of the rotor instead of the stator:

- No commutator needed.
- No carbon brushes.
- Reduced power loss due to friction.
- Higher speeds and generate higher torques.



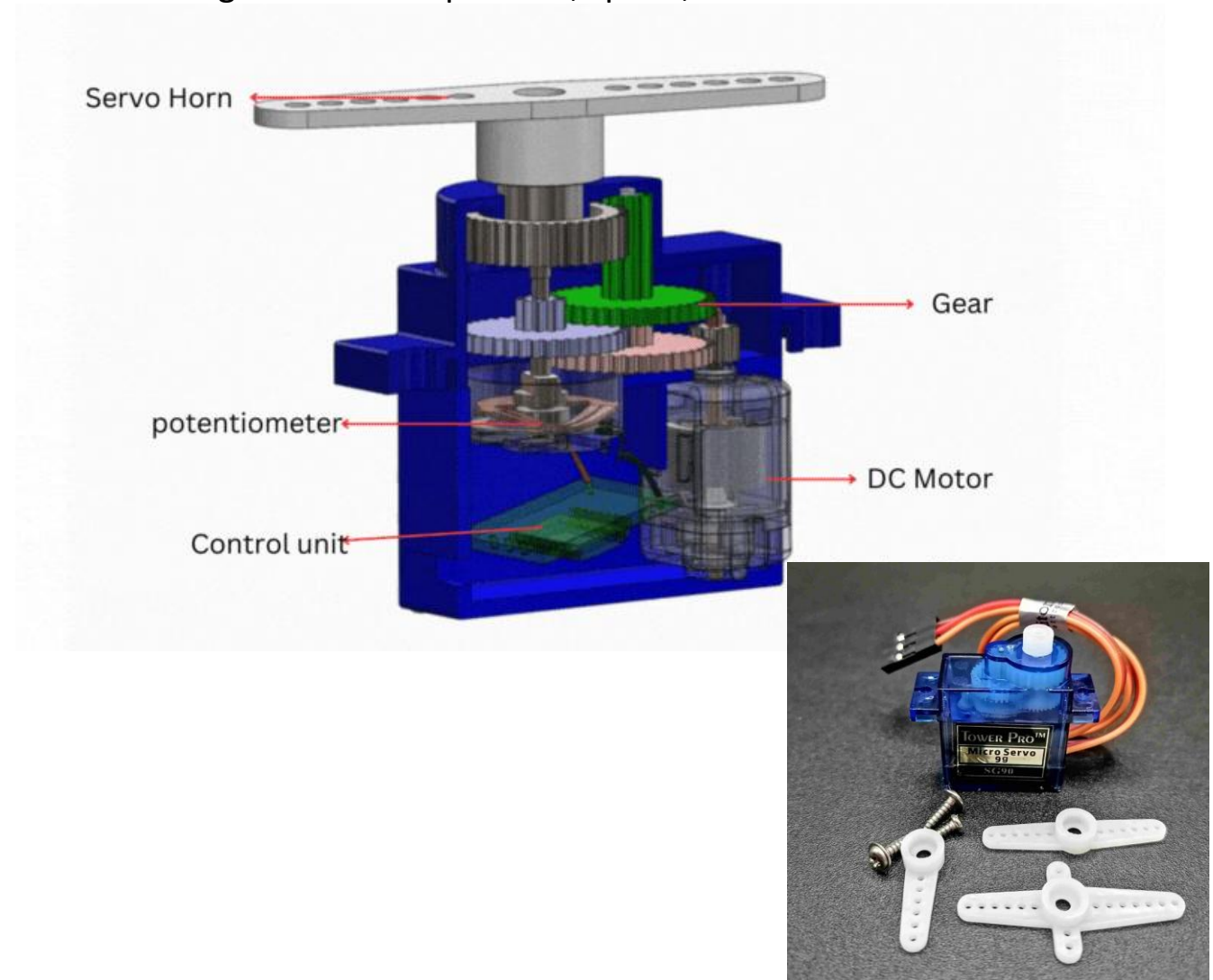
Servo motors

Work in a closed-loop configuration.

Servo motors use a feedback sensor to provide precise control over its angular or linear position, speed, and acceleration.

Key components:

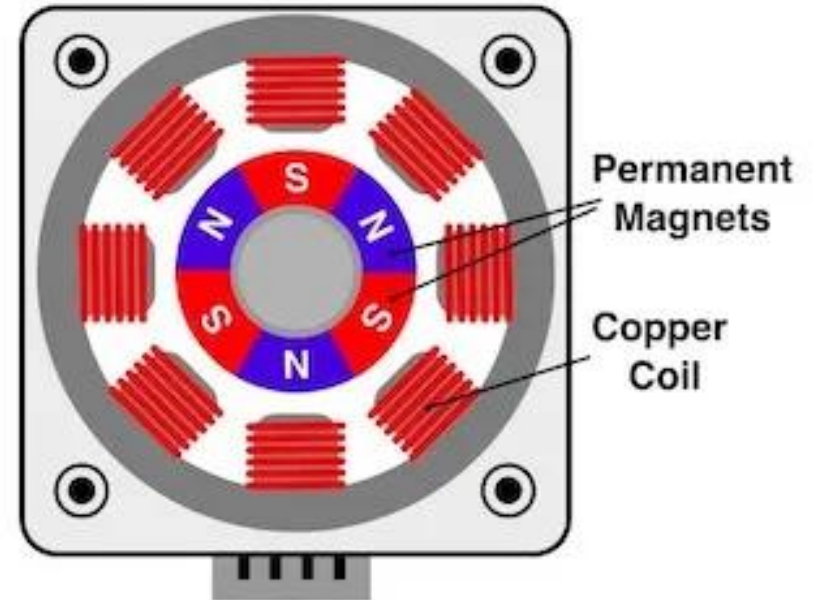
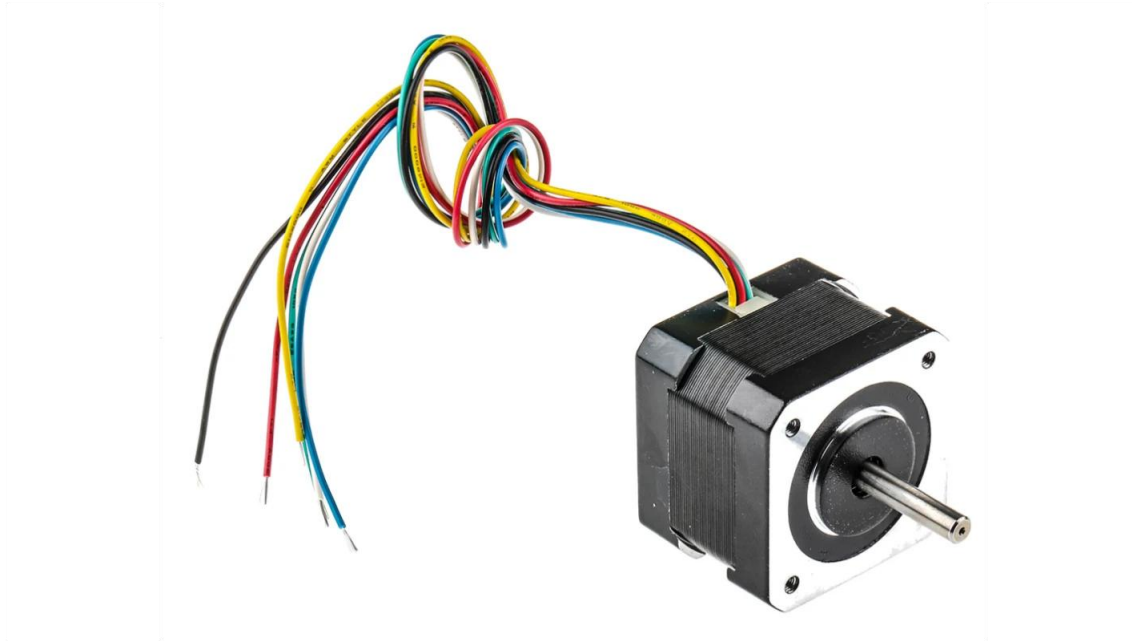
- **Motor:** The electromechanical device that provides torque and converts electrical energy into mechanical motion. This can be a DC motor, brushed or brushless.
- **Feedback sensor** (e.g., encoder or potentiometer attached to the motor shaft): It continuously monitors the motor's current position and provides real-time feedback to the control circuit.
- **Control circuit:** It compares the commanded position from the input signal with the actual position reported by the feedback sensor.
- **Gear assembly:** Attached to the motor shaft, this gearing reduces the motor's high speed to increase the output torque and improve precision.



Stepper motors

Stepper motors also keep track of rotor position, but indirectly.

Stepper motors rely on **knowing the number of steps** the rotor must move through to position it (incorporating a lot more incremental steps than “normal” electric motors).



Permanent magnet stepper motor

Elements of a Robot

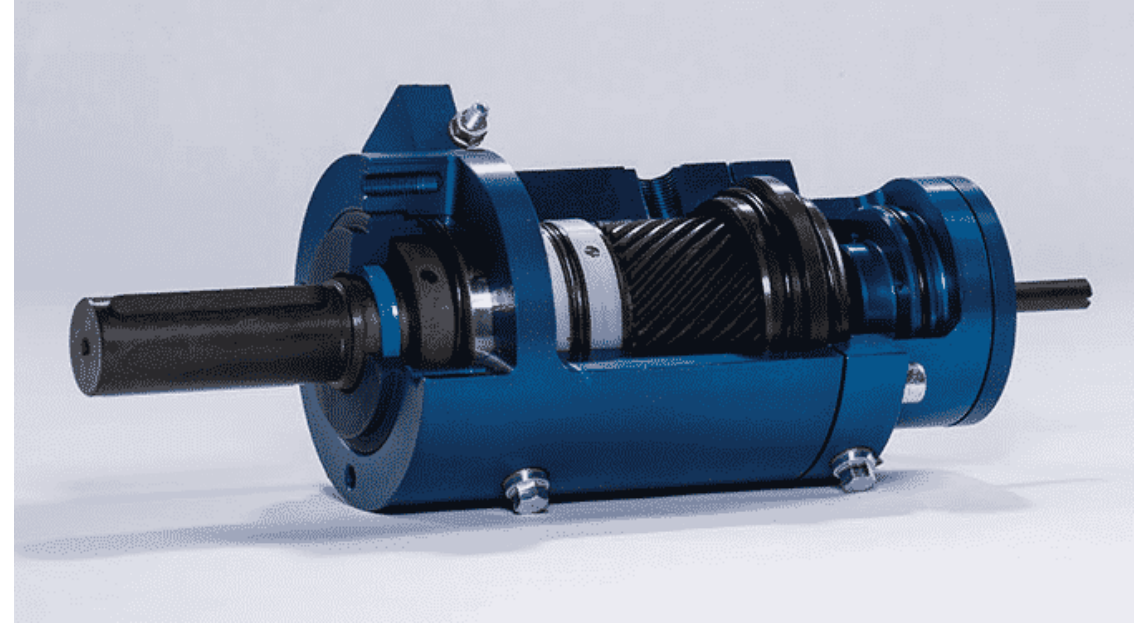
Hydraulic Actuators (~35% of cases)

Advantages:

- High load capacity.
- High speed.
- Position maintained due to oil incompressibility.
- Precise control possible.

Disadvantages:

- Relatively high cost for small sizes.
- Noisy and prone to oil leaks.
- Larger overall size.



Elements of a Robot

Pneumatic Actuators (~15% of cases)

Advantages:

- Relatively low cost.
- High speed.

Disadvantages:

- Limited accuracy (due to air compressibility).
- Noisy and prone to leaks.
- Require air filters and regular maintenance.

